

Lab 1: Math Refresher

Instructor notes – Part 1: prep reading, Part 2: podium/circulating cheat sheet

This single file has two parts, both instructor-only (not a student handout). **Part 1** is read beforehand – full worked examples and the reasoning behind the flow. **Part 2** (after the page break, large print) is the copy to have open while circulating during lab, since this session isn't one continuous lecture the way M/W periods are – it's more likely a short opening refresher followed by spot-checks with individual groups as they hit each Act.

Where this sits, and why it exists at all

Lab 1 (Tuesday, Week 1) needs log/exponent rules and a derivative (product rule + chain rule) in Act 2, and a first taste of integration in Act 2 Part 3 – **before any lecture has formally taught any of this**. Week 2 Lecture 1 (Monday) is where derivatives, critical points, and limits get properly developed; that lecture is *next week*. This doc is what fills the gap: a compressed, rule-focused refresher of material students are assumed to have seen before (Calc II-ish) but likely need dusted off, scoped tightly to what Act 2 actually asks for.

Deliberately excluded, per the “hold off on function analysis” call: no limits at infinity, no “throw out the lower-order terms,” no critical-point-vs-monotonic vocabulary, no inflection points. All of that is Week 2 Lecture 1's job, done properly with its own worked examples (Michaelis-Menten, logistic). Today is purely mechanical toolkit: the algebra and derivative rules needed to *execute* Act 2, not the conceptual apparatus for *characterizing* a function's shape.

Rationale for the flow, and an important design constraint

Order: exponents $\rightarrow$ logarithms (logs are defined in terms of exponents, so this order is closer to “necessary” than “chosen”) $\rightarrow$ power rule $\rightarrow$ product rule (both with geometric intuition, as requested) $\rightarrow$ chain rule $\rightarrow$ quotient rule (derived from product + chain, a forward-looking bonus since Act 2 doesn't need it) $\rightarrow$ a first taste of integration, closing with the Fundamental Theorem of Calculus.

Important: every worked numeric example below uses different numbers than Lab 1's actual questions. Act 2 gives students a specific log/exponent warm-up, a specific Ricker function ($a = 0.9$, $b = 1/12$) to differentiate, and a specific linear-ramp integration

problem – these are graded first-attempt exercises, not lecture content, and Act 2 already hands students the exact rule reminders they need (product rule, chain rule for $e^{g(x)}$) right on the page. Reusing Act 2’s actual numbers here would hand students the answer before they’ve tried it, undermining the whole point of the individual-work section. So: the *rules* and *techniques* below are identical to what Act 2 needs; the specific *numbers and functions* are all different (a different exponential decay, a different product-times-exponential function, a different integration ramp) – same skill, no spoilers. This is the same design choice made in Week 2 Lecture 1’s notes (Michaelis-Menten instead of Holling IV to teach a technique Lab 2 needed).

Exponent rules

Core rules (write these up as a reference block):

$$x^a \cdot x^b = x^{a+b} \quad \frac{x^a}{x^b} = x^{a-b} \quad (x^a)^b = x^{ab} \quad x^{-a} = \frac{1}{x^a}$$

Why $x^0 = 1$: don’t just assert it – use the pattern. Take $x = 2$ and divide by x each time you drop the exponent by one: $2^3, 2^2, 2^1, 2^0 = 8, 4, 2, 1$. Or, more rigorously, from the rule above: $x^a/x^a = x^{a-a} = x^0$, and anything divided by itself is 1 (for $x \neq 0$). Either argument is fine – the pattern is more intuitive, the algebra is more rigorous.

Worked examples (different numbers than Act 2’s):

$$2^x = 32 \Rightarrow x = 5 \quad 5^x = 125 \Rightarrow x = 3 \quad 10^x = 0.0001 \Rightarrow x = -4$$

For a base that isn’t a clean power, e.g. $e^x = 10$, you need logarithms – which is exactly the next section.

Logarithm rules

What a log actually is, before the rules: $\log_b(x)$ is the exponent you’d raise b to, to get x . This is worth saying plainly before any rule – students who can recite $\log(xy) = \log x + \log y$ often can’t say what a logarithm *is*.

Core rules:

$$\begin{aligned} \log(xy) &= \log x + \log y & \log(x/y) &= \log x - \log y & \log(x^k) &= k \log x \\ \log_b(b) &= 1 & \log_b(1) &= 0 & -\log(x) &= \log(1/x) \end{aligned}$$

The middle pair are just special cases of the exponent rules above (since $b^1 = b$ and $b^0 = 1$); the last one follows from $\log(x^{-1}) = -\log(x)$, itself a special case of $\log(x^k) = k \log x$ with $k = -1$.

Order of magnitude – why $\log_{10}$ is everywhere in science: for a number that’s an exact power of ten, $\log_{10}$ just counts the zeros: $\log_{10}(100) = 2$, $\log_{10}(1,000,000) = 6$. For

anything else, $\log_{10}$ gives you the *order of magnitude* – roughly how many digits the number has. E.g. $\log_{10}(7,000,000) \approx 6.85$: between 10^6 and 10^7 , i.e. “in the millions.” This is the entire reason log scales exist – they compress numbers spanning many orders of magnitude (population sizes, likelihoods, light intensity...) onto a scale where equal *distances* mean equal *multiplicative* factors.

Worked examples (different numbers than Act 2’s):

$$\log_{10}(2) + \log_{10}(50) = \log_{10}(100) = 2 \quad \log_{10}(500) - \log_{10}(5) = \log_{10}(100) = 2$$

$$\log_{10}(3^4) = 4 \log_{10}(3) \approx 4(0.477) = 1.91$$

Negative log = log of the reciprocal, concretely: $-\log_{10}(0.01) = \log_{10}(100) = 2$ – worth doing side by side with $\log_{10}(0.01) = -2$ directly, so students see both routes land on the same number.

Linearizing an exponential decay – the technique, generically. Given $y = ae^{-bx}$, take $\ln$ of both sides and apply the product and power-of-log rules:

$$\ln y = \ln(a e^{-bx}) = \ln a + \ln(e^{-bx}) = \ln a - bx$$

This is linear in x : slope $-b$, intercept $\ln a$. **Check with a numeric example** (again, different from Act 2’s $a = 10$, $b = 0.5$): $y = 5e^{-0.2x}$.

##	x	y	ln_y
## 1	0	5.00	1.609
## 2	5	1.84	0.609
## 3	10	0.68	-0.391
## 4	15	0.25	-1.391

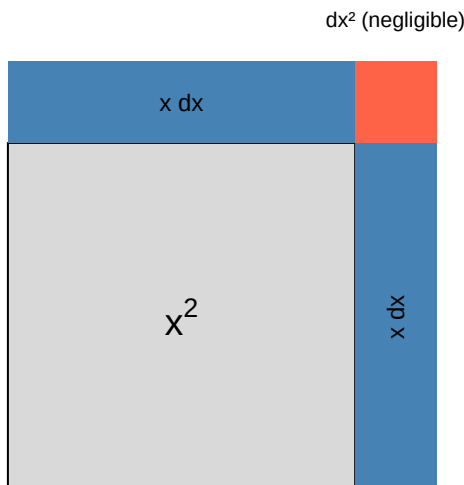
$\ln(y)$ drops by exactly 1.0 every time x increases by 5 – matching $-b\Delta x = -0.2 \times 5 = -1$ exactly, confirming the slope interpretation.

Power rule

Statement: $\frac{d}{dx}x^n = n x^{n-1}$.

Geometric intuition – x^2 as the area of a growing square. Picture a square of side x ; area x^2 . Grow the side by a small amount dx . The new area is the old square, plus a thin strip along the top ($x \times dx$), plus a thin strip along the right side ($x \times dx$), plus a tiny corner square ($dx \times dx = dx^2$) that’s negligible – it shrinks *faster* than the strips as $dx \rightarrow 0$ (it’s second-order small). So

$$\Delta(\text{Area}) \approx x dx + x dx = 2x dx \Rightarrow \frac{d}{dx}x^2 = 2x$$



The same picture generalizes: for x^3 (a cube), growing the side by dx adds three faces of area $x^2 dx$ each (plus even-smaller leftover terms), giving $3x^2 dx$ – matching $n x^{n-1}$ with $n = 3$.

Derivation from the product rule, by induction. $x^2 = x \cdot x$; product rule gives $1 \cdot x + x \cdot 1 = 2x$. $x^3 = x \cdot x^2$; product rule gives $1 \cdot x^2 + x \cdot (2x) = 3x^2$ (using the x^2 result just derived). In general, if $\frac{d}{dx}x^{n-1} = (n-1)x^{n-2}$ (assume this holds), then since $x^n = x \cdot x^{n-1}$:

$$\frac{d}{dx}x^n = 1 \cdot x^{n-1} + x \cdot (n-1)x^{n-2} = x^{n-1} + (n-1)x^{n-1} = n x^{n-1}$$

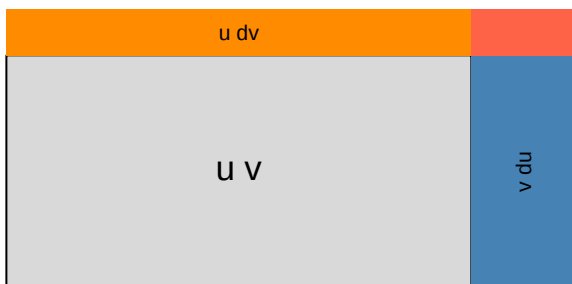
Each power's derivative is built from the previous one – the power rule is a *consequence* of the product rule, not a separate fact.

Product rule

Statement: $\frac{d}{dx}(uv) = u'v + uv'$.

Geometric intuition – a growing rectangle. Picture a rectangle with width $u(x)$ and height $v(x)$; area uv . As x changes by a small dx , u changes by $du = u'dx$ and v changes by $dv = v'dx$ – generally *different* small amounts, since u and v are different functions. The new area is the old rectangle, plus a strip along the right ($v \times du$), plus a strip along the top ($u \times dv$), plus a tiny corner ($du \times dv$) that's negligible for the same reason as before – a product of two small things is smaller still.

$$\Delta(\text{Area}) \approx v du + u dv \Rightarrow \frac{d}{dx}(uv) = u'v + uv'$$



Same picture as the power rule (unsurprising, since the power rule is a special case with $u = v = x$), just with the two sides allowed to grow at different rates.

Chain rule

Statement: $\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$, or in Leibniz notation, $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$ for $y = f(u)$, $u = g(x)$.

Intuition, not geometry this time – rates multiply through a composition. If you bike $3\times$ as fast as you walk, and you walk $2\times$ as fast as you crawl, then you bike $6\times$ as fast as you crawl – the rates *multiply*. That’s the whole idea: $f'(g(x))$ is the outer rate (how fast f changes per unit of u), $g'(x)$ is the inner rate (how fast u changes per unit of x), and the chain rule multiplies them to get the overall rate.

The one form Act 2 actually needs:

$$\frac{d}{dx}e^{g(x)} = g'(x) e^{g(x)}$$

(outer function $f(u) = e^u$ has $f'(u) = e^u$; multiply by the inner derivative $g'(x)$.)

Product + chain together – a worked example (not Ricker)

Act 2 asks for the derivative of $y = a x e^{-bx}$ using exactly this combination (product rule across the two factors, chain rule inside the exponential). Don’t work that example here – it’s the actual lab question. Instead, work a structurally identical warm-up:

$$f(x) = x^2 e^{-x}$$

```
# symbolic check via numerical differentiation, just to confirm the
# hand-derived answer below -- not shown to students, sanity check only
f <- function(x) x^2 * exp(-x)
fp_formula <- function(x) x * (2 - x) * exp(-x)
x0 <- 1.3
(f(x0 + 1e-6) - f(x0 - 1e-6)) / 2e-6 # numerical derivative
```

```
## [1] 0.2480039
```

```
fp_formula(x0) # hand-derived formula
```

```
## [1] 0.2480039
```

Product rule with $u = x^2$ ($u' = 2x$) and $v = e^{-x}$ ($v' = -e^{-x}$, chain rule with inner derivative -1):

$$f'(x) = 2x e^{-x} + x^2(-e^{-x}) = e^{-x}(2x - x^2) = x(2 - x)e^{-x}$$

Critical points at $x = 0$ and $x = 2$ (each factor that can be zero, set to zero) – exactly the “which factor could be zero” move Act 2’s Step 4 hints at, just demonstrated on a different function first.

Quotient rule – derived from product + chain (bonus, forward-looking)

Not needed for Act 2 (Ricker is a product, not a quotient), but next week’s bestiary is full of quotients (Michaelis-Menten, Holling II/III/IV) – worth showing it isn’t a fifth rule to memorize, it falls out of the two already in hand.

Write $\frac{u}{v} = u \cdot v^{-1}$. Apply the product rule:

$$\frac{d}{dx} (u \cdot v^{-1}) = u' \cdot v^{-1} + u \cdot \frac{d}{dx} (v^{-1})$$

The second piece needs the chain rule (outer function w^{-1} , power rule gives $\frac{d}{dw} w^{-1} = -w^{-2}$, times inner derivative v'):

$$\frac{d}{dx} (v^{-1}) = -v^{-2} v'$$

Substitute back and put over a common denominator:

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{u'}{v} - \frac{uv'}{v^2} = \frac{u'v - uv'}{v^2}$$

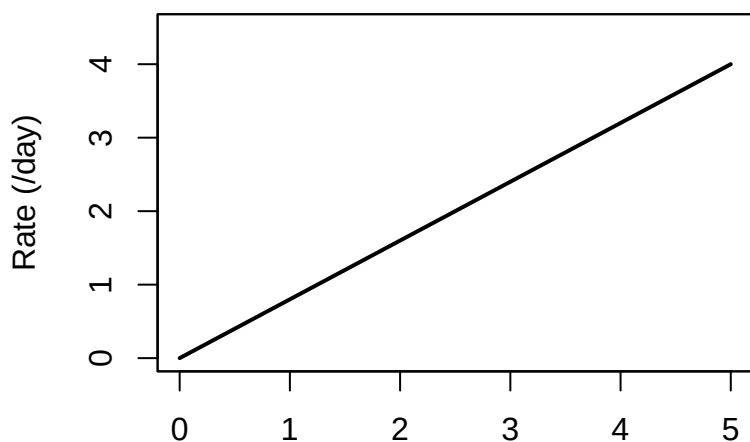
Every piece of the familiar quotient-rule formula is just product rule + chain rule + power rule, chained together. Worth saying explicitly: by the end of today they’ll have *all four* differentiation rules using only two real ideas (product rule, chain rule) plus the algebra trick $u/v = uv^{-1}$.

A first taste of integration

Two tools in calculus: derivatives (rate of change, from a total) and **integrals** (accumulation – total, from a rate). Act 2 Part 3 already gives students the simplest case fully worked (a constant capture rate) and asks them to handle a linear ramp themselves – so this section reuses their constant-rate example (safe, it’s exposition in the handout, not a graded task) and demonstrates the ramp technique on *different* numbers before they try their own.

Constant rate (Act 2’s own worked example, safe to reuse): rate $r = 2$ tadpoles/day for $T = 3$ days. Total $= r \times T = 6$ – the area of a 3×2 rectangle.

Ramping rate – worked here with different numbers than Act 2’s “your turn”: suppose the rate ramps linearly from 0 to 4 tadpoles/day over $T = 5$ days.



Total = area of the triangle $= \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2}(5)(4) = 10$.

Fundamental Theorem of Calculus, stated informally: if F is an antiderivative of f (i.e. $F' = f$), then the total accumulated change from a to b is

$$\int_a^b f(x) dx = F(b) - F(a)$$

– integration and differentiation are inverse operations. Check this against the triangle above using the power rule *in reverse*: the ramp is $f(t) = \frac{4}{5}t$, so an antiderivative is $F(t) = \frac{4}{5} \cdot \frac{t^2}{2} = \frac{2}{5}t^2$ (power rule backwards: raise the power by one, divide by the new power). Then

$$\int_0^5 \frac{4}{5}t dt = F(5) - F(0) = \frac{2}{5}(25) - 0 = 10$$

Matches the triangle’s area exactly – the geometric shortcut (area of a simple shape) and the general machinery (antiderivative, evaluated at the endpoints) agree, because the geometric shortcut *is* a special case of the general machinery. This is also a nice full-circle moment: today opened with the power rule, and it closes by running that same rule in reverse.

Optional aside, only if it lands naturally: just as a grid search (Week 1’s algorithms lecture) approximates an optimum by checking many points and gets more exact as you check

more of them, a Riemann sum approximates area by adding up many thin rectangles, and gets more exact as the rectangles get thinner. Both ideas are “approximate with many small pieces, then take a limit” – don’t force this connection if time is short, it’s a bonus thread, not required.

Take-home message

- Every rule below is something students have probably seen before; today’s job is to make it fluent again, not teach it for the first time.
- Power rule and product rule both come from the same geometric picture (a growing rectangle/square); quotient rule is product + chain rule plus one algebra trick ($u/v = uv^{-1}$); nothing here is a standalone fact to memorize in isolation.
- Every worked example above uses different numbers than Lab 1’s actual questions – if a group asks “is this like the example you showed us,” the honest answer is “same technique, different numbers, you still have to do the algebra.”

How to use this during lab

This isn’t a single continuous lecture – likely a short (10-15 min) opening pass through the high points (exponent/log rules, product + chain rule, one FTC example), then this doc becomes a reference while circulating. Quick map of what each Act needs, for triage while walking the room:

- **Act 2, Part 1** (log/exponent warm-up): exponent rules, log rules, linearizing an exponential decay.
- **Act 2, Part 2** (Ricker derivative + critical point): product rule, chain rule (for e^{-bx}), “set the derivative to zero and solve” – the x^2e^{-x} worked example above is the closest parallel to point a stuck group toward.
- **Act 2, Part 3** (integration taste): the rectangle/triangle picture, “area under a rate curve = total.”
- **Act 1 and Act 3** don’t need any of today’s math – they’re numeric/visual diagnosis and arithmetic (SSR), respectively. Don’t spend refresher time there.

PART 2 – CHEAT SHEET (circulating / opening pass)

Copy for the podium or for circulating during lab. Full derivations and rationale: see Part 1 above.

Exponent rules

$$x^a x^b = x^{a+b} \quad \frac{x^a}{x^b} = x^{a-b} \quad (x^a)^b = x^{ab} \quad x^{-a} = \frac{1}{x^a} \quad x^0 = 1$$

- $x^0 = 1$: pattern (x^3, x^2, x^1, x^0) , dividing by x each step) or $x^a/x^a = x^{a-a} = x^0 = 1$
- $2^x = 32 \Rightarrow x = 5$; $10^x = 0.0001 \Rightarrow x = -4$

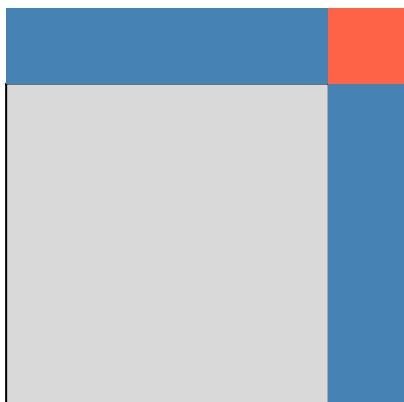
Logarithm rules

$$\log(xy) = \log x + \log y \quad \log(x/y) = \log x - \log y \quad \log(x^k) = k \log x \quad -\log x = \log(1/x)$$

- $\log_{10}$ = order of magnitude: $\log_{10}(1,000,000) = 6$ (six zeros); non-round numbers land in between
- linearizing decay: $y = ae^{-bx} \Rightarrow \ln y = \ln a - bx$ (line, slope $-b$, intercept $\ln a$)

Power rule – growing square

$$\frac{d}{dx}x^n = n x^{n-1}$$



- two strips ($x dx$ each) + negligible corner (dx^2) $\rightarrow 2x dx$
- from product rule by induction: $x^n = x \cdot x^{n-1} \rightarrow n x^{n-1}$

Product rule – growing rectangle

$$\frac{d}{dx}(uv) = u'v + uv'$$



- right strip $v du$ + top strip $u dv$ + negligible corner

Chain rule – rates multiply

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x) \qquad \frac{d}{dx}e^{g(x)} = g'(x)e^{g(x)}$$

- bike $3\times$ walk, walk $2\times$ crawl $\rightarrow$ bike $6\times$ crawl
- **warm-up example (not Ricker!):** $f(x) = x^2e^{-x} \rightarrow f'(x) = x(2 - x)e^{-x}$, critical pts $x = 0, 2$

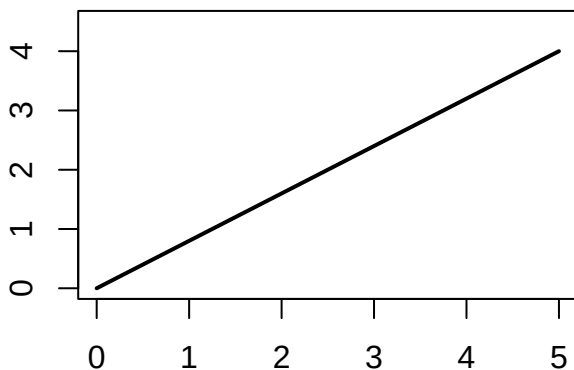
Quotient rule – product + chain, for free

$$\frac{u}{v} = uv^{-1} \rightarrow \frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$$

- no new idea – product rule + chain rule (v^{-1}) + power rule
- previews next week's quotient-shaped bestiary (MM, Holling)

A first taste of integration

- derivative = rate from total; **integral** = total from rate
- constant rate $r = 2$, $T = 3$ days $\rightarrow$ total $= rT = 6$ (rectangle)
- ramp $0 \rightarrow 4$ over 5 days $\rightarrow$ total $= \frac{1}{2}(5)(4) = 10$ (triangle)



- **FTC:** $\int_a^b f(x)dx = F(b) - F(a)$ where $F' = f$ (inverse ops)
- check: $f(t) = \frac{4}{5}t$, antiderivative $F(t) = \frac{2}{5}t^2$ (power rule in reverse)
 $\rightarrow F(5) - F(0) = 10$ – matches the triangle

Act-by-Act quick reference (for circulating)

- **Act 2 Pt.1** (log/exp warm-up) – exponent + log rules
- **Act 2 Pt.2** (Ricker derivative) – product rule + chain rule; point stuck groups at the x^2e^{-x} warm-up above, not the answer
- **Act 2 Pt.3** (integration taste) – rectangle/triangle, area = total
- **Act 1, Act 3** – no calculus needed, don't spend time here